

PARTS LIST

Semi-conductors:

IC1	- LM386 low-power audio amplifier
T1, T2	- BC549 npn transistor
LED1	- 5mm LED
Resistors (all 1/4-watt, $\pm 5\%$ carbon):	
R1, R6	- 4.7-kilo-ohm
R2, R3	- 2.2-kilo-ohm
R4	- 470-kilo-ohm
R5	- 1-kilo-ohm
R7	- 10-ohm
VR1	- 100-kilo-ohm potentiometer

Capacitors:

C1	- 0.01 μ F ceramic disk
C2	- 47pF ceramic disk
C4, C7	- 0.1 μ F ceramic disk
C5, C9	- 100 μ F, 25V electrolytic
C6	- 470 μ F, 25V electrolytic
C8	- 10 μ F, 25V electrolytic

Miscellaneous:

CON1	- 2-pin connector
BATT.1	- 12V battery
S1	- On/off switch
LS1	- 8-ohm, 0.5W speaker
	- 2-pin connector terminal for battery

close the PCB in a suitable box and place it near the guitar.

For construction of the guitar, take a wooden plank measuring approximately 10cm $\times$ 30cm and put four

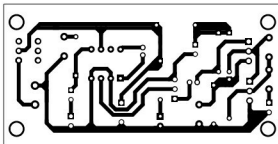


Fig. 4: Actual-size PCB of Faraday's guitar

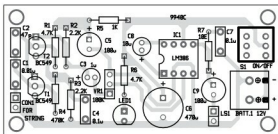


Fig. 5: Component layout of the PCB

nails on either side as shown in Fig. 1. Tie a copper wire (string) between the nails with correct tension. The copper wire can be taken out from any transformer with 500mA secondary rating.

Ensure that a single string is

passed between all the nails without joints or breaks. Place a magnet on the wooden plank roughly below the middle of the string and ensure the gap is small between the string and the magnet. The produced EMF depends on the air gap between the magnet and the string. If the magnet touches the string, it may obstruct the vibration of the string and the produced EMF would be low. If the gap is too much then, due to weak magnetic linkage, EMF will be lower. Therefore keep the magnet as close as possible to get higher EMF. ●



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